



The FPGA builds

Four FPGA builds that turn RTL into real hardware

This chapter is where your Verilog stops being “code” and becomes behaviour you can touch: LEDs that respond correctly, a PWM output that feels like control, a UART that prints meaningful debug, and an SPI link that proves you can talk to peripherals. Each project follows the same Fast Track template:

Spec → RTL blocks → Constraints → Build bitstream → Program → Debug → Upgrades → Portfolio

We’ll assume you’re using the open tool flow from Chapter 10: Yosys for synthesis and nextpnr for place-and-route (iCE40 via Project IceStorm, ECP5 via Project Trellis).

Before you start (common setup):

- **Create folders:** `src/`, `constraints/`, `build/`, `docs/`.

- **Grab your board’s pin map and identify:** `clk`, `led0`, `btn0` (and later UART/SPI pins).
- **Remember:** iCE40 uses PCF (`--pcf`), ECP5 uses LPF (`--lpf`).

Project 1: Blinky, but proper (with debounced button)

What you’ll build

- LED blinks at a steady rate (“heartbeat”)
- Button toggles between slow blink and fast blink
- Button is debounced so one press = one action

Why it matters: Most “my FPGA is haunted” moments are actually floating inputs or switch bounce. Mechanical bounce can generate multiple transitions for one press.